8 Feb 10: Subgradient Method Analysis and Projections

8.1 Subgradient Method Rates

Definition 8.1. *For the subgradient method for* $\min_{x\in\mathbb{R}^n} f(x)$ *and* f *convex where*

$$x_{t+1} = x_t - \eta_t g_t \text{ where } g_t \in \partial f(x_t).$$

We define the best iterate as

$$x_T^{(\text{best})} = \arg\min_{i=0,\dots,T} f(x_i).$$

Theorem 8.2. *Assume* f *is* G*-lipschitz and convex. Let* $||x_0 - x^*|| \le R$ *then pick*

$$\eta_t = \frac{R}{G\sqrt{T}} \text{ guarantees that } f\left(x_T^{(\text{best})}\right) - f(x^*) \le \frac{GR}{\sqrt{T}}.$$

Theorem 8.3. *A convex function* f *is* G*-lipschitz iff*

$$||g_x|| \le G \ \forall x \in \text{dom}(f) \text{ and } \forall g_x \in \partial f(x).$$

Theorem 8.4. *For nonconvex, differentiable* f*,* f *being* G*-lipschitz iff*

$$||\nabla f(x)|| \le G \ \forall x \in \text{dom}(f).$$

Theorem 8.5. *Assume* f *is convex and* G*-Lipschitz, and that an optimal solution* x^* *exists with* $||x_0 - x^*|| \le R$ *for some* $R > 0$*.*
Pick $\eta_t \to 0$ *such that* $\sum_{t=1}^{\infty} \eta_t = \infty$ *and* $\sum_{t=1}^{T} \eta_t^2 < \infty$ *then*

$$f\left(x_T^{(\text{best})}\right) \to f(x^*) \text{ as } T \to \infty.$$

Theorem 8.6. *For the subgradient method with step sizes* $\{\eta_t\}_{t=1}^T$ *on a convex,* G*-Lipschitz function* f*, we have*

$$f\left(x_T^{(\text{best})}\right) - f(x^*) \ \le \ \frac{\|x_0 - x^*\|^2 + G^2 \sum_{t=1}^T \eta_t^2}{2\sum_{t=1}^T \eta_t}.$$

Verification of Theorem 8.2. Set $\eta_t = \frac{R}{G\sqrt{T}}$ for all t. With $\|x_0 - x^*\| \le R$:

$$\sum_{t=1}^{T} \eta_t = T \cdot \frac{R}{G\sqrt{T}} = \frac{R\sqrt{T}}{G},$$
$$\sum_{t=1}^{T} \eta_t^2 = T \cdot \frac{R^2}{G^2 T} = \frac{R^2}{G^2}.$$

Substituting into the bound (using $\|x_0 - x^*\|^2 \le R^2$):

$$f\left(x_T^{(\text{best})}\right) - f(x^*) \le \frac{R^2 + G^2 \cdot \frac{R^2}{G^2}}{2 \cdot \frac{R\sqrt{T}}{G}} = \frac{2R^2}{2R\sqrt{T}/G} = \frac{2R^2 \cdot G}{2R\sqrt{T}} = \frac{GR}{\sqrt{T}}.$$

Hence Theorem 8.2 holds with the $O(1/\sqrt{T})$ rate. □

Proof.

$$\begin{aligned} ||x_{t+1} - x^*||^2 &= ||x_t - \eta_t g_t - x^*||^2 \\ &= ||x_t - x^*||^2 - 2\eta_t g_t^T (x_t - x^*) + \eta_t^2 ||g_t||^2 \\ &\le ||x_t - x^*||^2 + 2\eta_t (f(x^*) - f(x_t)) + \eta_t^2 G^2 \qquad (*) \end{aligned}$$

Last step follows from $f(x^*) \geq f(x_t) + g_t^T(x^* - x_t) \iff -g_t^T(x_t - x^*) \geq f(x^*) - f(x_t)$.
Adding $(*)$ up from $0, ..., T-1$ we have

$$||x_T - x^*||^2 \leq ||x_0 - x^*||^2 + G^2 \sum_{t=0}^{T-1} \eta_t^2 + 2 \sum_{t=0}^{T-1} \eta_t \left(f(x^*) - f(x_t)\right).$$

So

$$2 \sum_{t=0}^{T-1} \eta_t \left(f(x_T^{(\text{best})}) - f(x^*)\right) \leq 2 \sum_{t=0}^{T-1} \eta_t \left(f(x_t) - f(x^*)\right) \leq ||x_0 - x^*||^2 + G^2 \sum_{t=0}^{T-1} \eta_t^2.$$

□

8.2 Polyak Stepsize and Feasibility

Theorem 8.7. *Polyak's Stepsize: If $f(x^*)$ is known,*

$$\eta_t = \frac{f(x_t) - f(x^*)}{||g_t||^2}.$$

Theorem 8.8. *Given $C_1, ..., C_k$ convex sets find $x^* \in \bigcap_{i=1}^k C_i$.*

Proof. Define $f_i(x) = \min_{y \in C_i} ||x - y||^2 = \text{dist}(x, C_i)$ and $f(x) = \max_{i=1,...,k} f_i(x)$.
If $x^* \in C_1 \cap ... \cap C_k$ then $f(x^*) = 0$.
Recall: $\partial f(x) = \text{Conv}\left(\bigcup_{i=1,...,k} \partial f_i(x)\right)$.
Let $P_C(x) = \arg\min_{y \in C} ||x - y||$

Lemma 8.9. *u is the projection of x onto C iff $\langle x - u, y - u \rangle \leq 0 \forall y \in C$*

We have $f_i(x) = ||x - P_{C_i}(x)||^2$ so $\partial f_i(x) = \frac{x - P_{C_i}(x)}{|x - P_{C_i}(x)||}$ if $x \neq 0$.

$$x_{t+1} = x_t - f(x_t) \cdot \frac{x_t - P_{C_i}(x_t)}{||x_t - P_{C_i}(x_t)||} = P_{C_i}(x_t).$$

□

9 Feb 12: Projected Subgradient Method

9.1 Projections and Contractions

Definition 9.1. *For* $\min_{x\in C} f(x)$ *we define the projected subgradient method to be*

$$\begin{cases} y_{t+1} = x_t - \eta_t g_t \\ x_{t+1} = P_C(y_t) \end{cases}.$$

Theorem 9.2. $P_C(x) = \arg\min_{y\in C} ||x-y||^2$

Proof. We have $z = P_c(x)$ iff $\forall y \in C$, $\langle x-z, y-z\rangle \leq 0$. Then for

$$\underbrace{-\nabla f(z)}_{x-z} \in \underbrace{N_c(z)}_{\{g:g^T(y-z)\leq 0\ \forall y\in C\}}.$$

so we have $(x-z)^T(y-z) \leq 0$ □

Lemma 9.3. *Projections are contractions:*

$$||\underbrace{P_C(x_1)}_{z_1} - \underbrace{P_C(x_2)}_{z_2}||_2 \leq ||x_1 - x_2||_2.$$

Proof. We have

1. $\langle x_1 - z_1, z_2 - z_1\rangle \leq 0 \forall z_2 \in C$
2. $\langle x_2 - z_2, z_1 - z_2\rangle \leq 0 \forall z_1 \in C$
3. Adding these two inequalities we have

$$\langle x_1 - z_1, z_2 - z_1\rangle + \langle x_2 - z_2, z_1 - z_2\rangle = \langle x_1 - x_2 + z_2 - z_1, z_2 - z_1\rangle \leq 0.$$

So

$$\langle x_1 - x_2, z_2 - z_1\rangle + ||z_2 - z_1||_2^2 \leq 0 \iff ||z_2 - z_1||_2^2 \leq \langle x_1 - x_2, z_2 - z_1\rangle \leq ||x_1 - x_2||||z_2 - z_1||.$$

So we have our result

$$\|z_1 - z_2\| \leq ||x_1 - x_2||.$$

□

9.2 Convergence and Examples

Theorem 9.4. *Rates for PGD are identical to GD and similarly PSGM is identical to SGM.*

Proof.

$$||x_{t+1} - x^*||^2 = ||P_C(x_t - \eta_t g_t) - x^*||^2 = ||P_C(x_t - \eta_t g_t) - P_C(x^*)|| \leq ||x_t - \eta_t g_t - x^*||^2.$$

□

Example 9.5. *Consider*

$$\min_{x\in\mathbb{R}^d} \frac{1}{2}||Ax - b||_2^2 \text{ where } n \geq d \text{ and } A \text{ full rank}.$$

When we solve this by gradient descent then the rate would be

$$\left(1 - \frac{\lambda_{min}}{\lambda_{max}}\right)^k.$$

Proof. We have $\nabla f(x) = A^T(Ax - b)$ and $\nabla^2 f(x) = A^T A$. So f is quadratic with Hessian $A^T A$. The eigenvalues of $A^T A$ satisfy $\lambda_{\min} = \lambda_{\min}(A^T A)$ and $\lambda_{\max} = \lambda_{\max}(A^T A)$. Hence f is α-strongly convex with $\alpha = \lambda_{\min}$ and β-smooth with $\beta = \lambda_{\max}$. Taking $\gamma = \beta/\alpha = \lambda_{\max}/\lambda_{\min}$ and $\eta = 1/\beta$, the strongly convex convergence theorem gives

$$||x_k - x^*||^2 \leq \left(1 - \frac{1}{\gamma}\right)^k ||x_0 - x^*||^2 = \left(1 - \frac{\lambda_{\min}}{\lambda_{\max}}\right)^k ||x_0 - x^*||^2.$$

□

Example 9.6. *Consider a similar problem*

$$\min_{x \geq 0} \frac{1}{2}||Ax - b||_2^2 \text{ where } n \geq d \text{ and } A \text{ full rank }.$$

We can let $y_{t+1} \leftarrow x_t - \eta_t(Ax_t - b)$ *and* $x_{t+1} \leftarrow \max(0, y_{t+1})$.

Example 9.7. *Example of Fast Projection:*

$$C = \left\{Ay + b | y \in \mathbb{R}^\ell\right\} \subseteq \mathbb{R}^n \text{ and } P_C(x) = b + A(A^T A)^{-1} A^T (x - b).$$

Proof.

$$y^* = \arg\min_{y \in \mathbb{R}^\ell} \frac{1}{2}||x - (Ay + b)||_2^2 \implies A^T(Ay^* + b) - x = 0 \implies A^T A y^* = A^T(x - b) \implies y^* = (A^T A)^{-1} A^T (x - b).$$

So $P_C(x) = Ay^* + b$ □

Example 9.8.

$$C = \{y : Ay = b\} \subseteq R^d \text{ and } P_c(x) = x + A^T(AA^T)^{-1}(b - Ax).$$

10 Feb 14: Optimality Conditions and Proximal Gradient Descent

Theorem 10.1. *Consider* $\min_{x\in\mathbb{R}^d} f(x)$, *$f$ not assumed convex. Then x^* is optimal on* $\mathrm{dom}(f)$ *iff* $0 \in \partial f(x^*)$.

Proof. ($\Longleftarrow$) We have

$$f(y) \geq f(x^*) + \langle g_{x^*}, y - x^* \rangle . \qquad (\forall y \in \mathrm{dom}(f) \text{ and } g_x^* \in \partial f(x^*))$$

Since $0 \in \partial f(x^*)$ we have $f(y) \geq f(x^*)$ for all $y \in \mathrm{dom}(f)$. So x^* is optimal.
($\Longrightarrow$)

$$f(y) \geq f(x^*) + \langle 0, y - x^*, \Longrightarrow \rangle 0 \in \partial f(x^*).$$

□

Theorem 10.2. *f is convex and differentiable with C convex then*

$$x^* \in \arg\min_{x\in C} f(x) \iff -\nabla f(x^*) \in N_C(x^*).$$

Theorem 10.3. *If f is convex and C convex then*

$$x^* \text{ is optimal for } f \text{ in } C \iff 0 \in \partial f(x^*) + N_C(x^*).$$

Proof. We can rewrite the objective as

$$x^* \in \arg\min_{x\in C} f(x) = \arg\min_{x\in\mathbb{R}^d} f(x) + I_C(x) \iff 0 \in \partial f(x^*) + N_C(x^*).$$

□

Example 10.4. *Soft Thresholding:*

$$x^* = \arg\min_{z\in\mathbb{R}^d} \frac{1}{2}||z - y||_2^2 + \lambda||z||_1.$$

We can conclude that

$$x^* = \begin{cases} y & \text{if } x^* = 0 \\ 0 & \text{if } x^* = \infty \end{cases}.$$

We have $0 \in x^* - y + \lambda sign(x^*)$. *So*

$$x_j^* = \begin{cases} y_j - \lambda & \text{if } y_j \geq \lambda \\ y_j + \lambda & \text{if } y_j \leq -\lambda \\ 0 & \text{otherwise} \end{cases}.$$

Theorem 10.5. *Proxiaml Gradient Descent: Our objective is*

$$\min_{x\in\mathbb{R}^d} \underbrace{g(x)}_{\substack{convex \\ diff}} + \underbrace{h(x)}_{\substack{convex \\ non-diff \\ simple}}.$$

Our update rules are

$$y^{t+1} \leftarrow x^t - \eta_t \nabla g(x^t)$$
$$x^{t+1} \leftarrow prox_{\eta_t,h}(y^{t+1}) := \arg\min_{z\in\mathbb{R}^d} \frac{1}{2\eta_t}||y^t - z||_2^2 + h(z)$$

When $g(x) = ||Ax - b||_2^2$, $h(x) = \lambda||x||_1$ *then this is Proximal GD called ISTA.*

Remark 10.6. *The intuition is that*

$$x_{t+1} \leftarrow \arg\min_z g(x_t)+\nabla g(x_t)^T(z-x_t)+\frac{1}{2\eta_t}||z-z_t||_2^2+h(z) = \frac{\arg\min_z 1}{2\eta_t}||z-(x_t-\eta_t\nabla g(x_t))||_2^2+h(z).$$

Theorem 10.7. *For* $p(x) = Prox_{\eta,h}(x) = \frac{\arg\min_z 1}{2\eta}||x-z||_2^2 + h(z)$ *we have*

1. $||p(x)-p(y)||_2^2 \leq \langle x-y, p(x)-p(y)\rangle$
2. $||p(x)-p(y)|| \leq ||x-y||$

(1) $\implies$ (2) *by the Cauchy-Schwarz inequality.*

Proof. We have

$$\frac{1}{\eta}(x-p(x)) \in \partial h(p(x)) \text{ and } \frac{1}{\eta}(y-p(y)) \in \partial h(p(y)).$$

By subgradient monotonicity we have

$$\langle g_x - g_y, x-y\rangle \geq 0 \text{ for all } x, y \text{ and } g_x \in \partial f(x), g_y \in \partial f(y).$$

So

$$\left\langle \frac{1}{\eta}(y-p(y)) - \frac{1}{\eta}(x-p(x)), p(y)-p(x)\right\rangle \geq 0 \iff \langle y-x, p(y)-p(x)\rangle \geq \langle y-x, p(y)-p(x)\rangle \geq ||p(y)-p(x)||_2^2.$$

□

Theorem 10.8. *Define* $G_\eta(x) = \frac{x-prox_{\eta,h}(x-\eta\nabla g(x))}{\eta}$ *then the following are true:*

1. $G_\eta(x) = 0 \iff x = prox_{\eta,h}(x-\eta\nabla g(x))$
2. $G_\eta(x^*) = 0 \iff 0 \in \nabla g(x^*) + \partial h(x^*) \iff x^*$ *is optimal* $\iff x^*$ *is a fixed point of prox*

Proof. By definition $\forall \tilde{x}$ then

$$\tilde{x} - \eta G_\eta(\tilde{x}) = prox_{\eta,h}(\underbrace{\tilde{x}-\eta\nabla g(\tilde{x})}_{x}).$$

$$\begin{aligned} u := prox_{\eta,h}(x) &\iff \frac{1}{\eta}(x-u) \in \partial h(u) \\ &\iff \tilde{x}-\eta\nabla g(\tilde{x}) - \tilde{x} - \eta G_\eta(\tilde{x}) \in \eta\partial h(\tilde{x}-\eta G_\eta(\tilde{x})) \\ &\iff G_\eta(\tilde{x}) \in \nabla g(\tilde{x}) + \partial h(\tilde{x}-\eta G_\eta(\tilde{x}))\ \forall \tilde{x} \end{aligned}$$

Thus we imply 1 by choosing $\tilde{x} = x^*$ □